

USER INPUT
*Enter expense amount
*select category
*Set monthly budget
*Click Submit button

FRONTEND UI
(HTML,CSS,Javascript)
*Form validation
*check empty fields
*Format input
*Display temporary data

API Request-JSON

BACKEND SERVER(Flask)
*Receive request(POST)
*Parse JSON data
*Validate data

IS INPUT VALID?

Process Data
*Add expense
*Update total

Send ERROR response
Invalid Input

DATABASE(SQLite)
*Stores expense record
*Store budget info
*Retrieve previous data

FRONTEND UI
Show error

BACKEND PROCESSING LOGIC
*Calculate total spending
*Compare with budget
*Calculate remaining budget

Budget Exceeded?

GENERATE ALERT
Budget exceeded

NORMAL STATUS
Within limit

RESPONSE(JSON)
*Total expenses
*Remaining budget
*Alert message
*Chart data

DATA ANALYSIS(Pandas)
*Weekly spending analysis
*Monthly trends
*Basic prediction

FRONTEND OUTPUT
*Display total expenses
*Show charts(graphs)
*Show alerts
*Update dashboard

User View(FINAL)
*Dashboard
*Expense Summary
*Budget status